

Lecture 7. kernels (SVM continued...)

outline: - Optimization problem

- Representer theorem for very high dimensional features

- kernels mechanism to work with high dimensional features

- Example of kernels

without explicit computation/design of ϕ

Recap:

Optimal margin classifier

$$\min_{w, b} \frac{1}{2} \|w\|^2 \text{ s.t. } y^{(i)}(w^T x^{(i)} + b) \geq 1 \quad (i=1, \dots, n)$$

$$\gamma^{(i)} = \frac{y^{(i)}(w^T x^{(i)} + b)}{\|w\|}$$

$$\gamma = \min_{i=1 \sim n} \gamma^{(i)}$$

(공의 양)

클래스 방향 맞춰서 거리 계산 ($y^{(i)}$ 와 $w^T x^{(i)} + b$ 의 부호가 같으면 부류 성공)

Decision boundary로부터 적어도 거리가 1 이상 떨어져 있도록

Optimization problem:

$$\max_{\gamma, w, b} \gamma \text{ s.t.}$$

$$\frac{y^{(i)}(w^T x^{(i)} + b)}{\|w\|} \geq \gamma$$

$$\geq \gamma$$

Geometric Margin

모든 샘플이 적어도 margin γ 이상 떨어져 있도록.

worst training example's GM을 lift up!
⇒ then every examples will have bigger GM

$$\begin{aligned} \text{e.g. } y &= g(w^T x + b) \\ &= g\left(\begin{bmatrix} 2 \\ 1 \end{bmatrix}^T x + 2\right) \\ &= g\left(\begin{bmatrix} 2 \\ 0 \end{bmatrix}^T x + 20\right) \end{aligned}$$

$$\text{sigmoid 함수 } g(x) = \frac{1}{1 + e^{-x}} \text{ 은 } x=0 \text{ 일때 } g(x)=0.5$$

$$\therefore w^T x + b = 0 \text{ 이 곧 decision boundary}$$

$$2x + x + 2 = 0 \text{ 와 } 20x + 0x + 20 = 0 \text{ 은 같다. } y \text{ 값이 달라도 classification은 같음}$$

$$\max \frac{1}{\|w\|} \text{ s.t. } y^{(i)}(w^T x^{(i)} + b) \geq 1$$

$$\min_{w, b} \frac{1}{2} \|w\|^2 \text{ s.t. } y^{(i)}(w^T x^{(i)} + b) \geq 1 \quad (i=1 \sim m)$$

수학적 편의성을 위하여 $\|w\|$ 대신 $\frac{1}{2}\|w\|^2$

Representation theorem:

$$x^{(i)} \in \mathbb{R}^{100}$$

$$\text{suppose } w = \sum_{i=1}^m \alpha_i y^{(i)} x^{(i)}$$

why? Intuition #1. Logistic regression

$$\theta := 0$$

$$\text{Gradient descent } \theta := \theta - \alpha (h_{\theta}(x^{(i)}) - y^{(i)}) \cdot x^{(i)}$$

$$(\text{batch GD}) \theta := \theta - \alpha \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) \cdot x^{(i)}$$

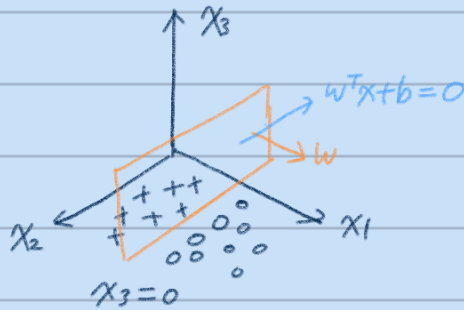
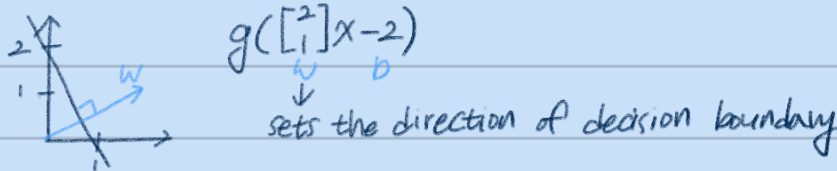
dual parameter: 각 훈련 샘플이 최적해에 얼마나 가까운지를 나타내는 변수

$\alpha_i = 0$ 이면 샘플이 최적해와 떨어져 영향을 주지 않음

$\alpha_i > 0$ 이면 $x^{(i)}$ 는 support vector가 됨

각 샘플 $x^{(i)}$ 에 비례하는 형태로 θ 가 업데이트

Intuition #2. 결정 경계의 방향은 결국 어떤 샘플들로부터 유도되나? ← 인스턴스



Assume $w = \sum_{i=1}^m \alpha_i y^{(i)} x^{(i)}$

plug into optimization problem: $\min_{w,b} \frac{1}{2} \|w\|^2$ s.t. $y^{(i)}(w^T x^{(i)} + b) \geq 1 \quad (i=1 \dots m)$

$$\min \frac{1}{2} \left(\sum_{i=1}^m \alpha_i y^{(i)} x^{(i)} \right)^T \left(\sum_{j=1}^m \alpha_j y^{(j)} x^{(j)} \right)$$

$$y^{(i)} \left(\left(\sum_j \alpha_j y^{(j)} x^{(j)} \right)^T x^{(i)} + b \right) \geq 1$$

$$= \min \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j y^{(i)} y^{(j)} \underbrace{x^{(i)T} x^{(j)}}_{\langle x^{(i)}, x^{(j)} \rangle}$$

$$y^{(i)} \left(\sum_j \alpha_j y^{(j)} \langle x^{(j)}, x^{(i)} \rangle + b \right) \geq 1$$

$\langle x, z \rangle$: $x^T z$ is the inner product *결국 내적으로 표현됨*

↓ ∴ convex optimization theory

Dual optimization problem

$$\max \sum_i \alpha_i - \frac{1}{2} \sum_i \sum_j y^{(i)} y^{(j)} \alpha_i \alpha_j \langle x^{(i)}, x^{(j)} \rangle \quad \text{s.t. } \alpha_i > 0 \quad \sum_i y^{(i)} \alpha_i = 0$$

1) Solve for α_i, b

2) Make a prediction $h_{w,b}(x) = g(w^T x + b)$

$$= g\left(\left(\sum_i \alpha_i y^{(i)} x^{(i)}\right)^T x + b\right)$$

$$= g\left(\sum_i \alpha_i y^{(i)} \langle x^{(i)}, x \rangle + b\right)$$

학습중 결정 (α 어떤 샘플은 support vector b 관련 (margin 위치))

SVM 최종 예측 함수 (학습된)

Kernel trick

1. Write algorithm in terms of $\langle x^{(i)}, x^{(j)} \rangle$ (or $\langle x, z \rangle$)
2. Let there be some mapping from $x \rightarrow \phi(x) = \begin{bmatrix} x \\ x^2 \\ x^3 \\ x^4 \end{bmatrix}$
3. Find way to compute $K(x, z) = \phi(x)^T \phi(z)$
4. Replace $\langle x, z \rangle$ in algorithm with $K(x, z)$

$K(x, z)$

원래 문제	margin을 키우기 위해 w 를 줄이자는 목적
Dual 변환	w가 $\alpha_i, x^{(i)}$ 의 조합이 됨
결정 함수	$f(x) = \sum \alpha_i y^{(i)} \langle x^{(i)}, x \rangle + b \leftarrow$ "유사도 기반" 표현이 등장
커널 도입	유사도 계산을 더 유연하게 하기 위해 $\langle x^{(i)}, x \rangle$ 을 $K(x^{(i)}, x)$ 로 확장
즉:	
graphical margin을 최대화하는 목적은 그대로 유지하면서, 이 목적을 더 유연하고 강력하게 구현하기 위해 커널로 표현이 바뀐 것이예요.	
비유로 말하자면	
• 처음에는 자와 컴퍼스를 들고 ***가장 넓은 간격의 선형 경계***를 찾고 있었는데, • 최적화 과정을 뒤집어 보니까 → "결정 경계는 결국 데이터와 데이터 간 유사도에 의해 정해지더라" • 그래서 자 대신 **유사도 측정 도구(Kernel)**로 바뀐 것!	

Example of Kernels

$$x \in \mathbb{R}^n \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad \phi(x) \in \mathbb{R}^{n^2} \begin{bmatrix} x_1 x_1 \\ x_1 x_2 \\ x_1 x_3 \\ x_2 x_1 \\ x_2 x_2 \\ x_2 x_3 \\ x_3 x_1 \\ x_3 x_2 \\ x_3 x_3 \end{bmatrix} \quad \phi(z) = \begin{bmatrix} z_1 z_1 \\ z_1 z_2 \\ \vdots \\ z_3 z_3 \end{bmatrix}$$

$\sqrt{2} \times \begin{matrix} x_1 \\ x_2 \\ x_3 \end{matrix}$
 $\begin{matrix} \sqrt{2} \times z_1 \\ \sqrt{2} \times z_2 \\ \sqrt{2} \times z_3 \end{matrix}$

$\left\{ \begin{array}{l} n^2 \text{ elements needs } O(n^2) \text{ time} \\ \text{to compute } \phi(x), \text{ or } \phi(x)^T \phi(z) \\ \text{explicitly} \end{array} \right.$

$$K(x, z) = \phi(x)^T \phi(z) = (x^T z)^2$$

$\uparrow \quad \uparrow$
 $\mathbb{R}^n \quad \mathbb{R}^n \quad O(n) \text{ time}$

$$= \left(\sum_{i=1}^n x_i z_i \right) \left(\sum_{i=1}^n x_i z_i \right)$$

$$= \sum_{i=1}^n \sum_{j=1}^n x_i z_i x_j z_j$$

$$K(x, z) = (x^T z + c)^2 \quad c \in \mathbb{R} \text{ (constant)}$$

$$K(x, z) = (x^T z + c)^d \quad O(n) \text{ time}$$

$\phi(x)$ has all $\binom{n+d}{d} \approx (n+d)^d$ features of monomials up to order d

d 가 커지면 computation time $O(n)$ 은 doesn't blow up

Support Vector Machine

1. Take the optimal margin classifier
 2. Apply the kernel trick to it $\rightarrow$ computation time scales only linearly
- $\Rightarrow$ forms linear decision boundary on high dimensional space

How to make kernels?

If x, z are "similar", $k(x, z) = \phi^T(x) \phi(z)$ is large

x, z are "dissimilar", $k(x, z)$ is small

$$k(x, z) = \exp\left(-\frac{\|x - z\|^2}{2\sigma^2}\right) \quad \text{Do only when there exists } \phi \text{ s.t. } k(x, z) = \phi(x)^T \phi(z)$$

Gaussian kernel $k(x, x) = \phi(x)^T \phi(x) \geq 0$

$\xrightarrow{\text{valid kernel}} \frac{309}{309}$ let $\{x^{(1)}, \dots, x^{(d)}\}$ be d points

let $K \in \mathbb{R}^{d \times d}$ "kernel matrix"

$$K_{ij} = k(x^{(i)}, x^{(j)})$$

Given any vector z ,

$$\begin{aligned} z^T K z &= \sum_i \sum_j z_i K_{ij} z_j \\ &= \sum_i \sum_j z_i \phi(x^{(i)})^T \phi(x^{(j)}) z_j \\ &= \sum_i \sum_j z_i \sum_k (\phi(x^{(i)})_k) (\phi(x^{(j)})_k) z_j \\ &= \sum_i \sum_j \sum_k z_i (\phi(x^{(i)})_k) (\phi(x^{(j)})_k) z_j \end{aligned}$$

$$= \sum_k \left(\sum_i z_i \phi(x^{(i)})_k \right)^2 \geq 0 \quad \text{So, } k \geq 0$$

Mercer's Theorem: k is a valid kernel function

$$(i.e. \exists \phi \text{ s.t. } k(x, z) = \phi(x)^T \phi(z))$$

if and only if for any d points $\{x^{(1)} \dots x^{(d)}\}$,

the corresponding kernel function matrix $K \geq 0$

Kernel의 종류

Linear kernel $k(x, z) = x^T z$ $\phi(x) = x$ 제일 많이 쓰임

Gaussian kernel $\phi(x) \in \mathbb{R}^\infty$

Polynomial kernel $k(x, z) = (x^T z + c)^d$

Expand kernel to $\langle x^{(i)}, x^{(i)} \rangle$

kernel is more general, it's not limited to SVM. Can be applied to other algorithms.

L1 Norm Soft Margin SVM keeps outliers as outliers (overfitting 방지)

$$\min_{w, b, \xi_i} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^m \xi_i \quad \text{s.t.} \quad \underbrace{y^{(i)}(w^T x^{(i)} + b)}_{\text{functional margin}} \geq 1 - \xi_i \quad (i=1, \dots, m) \quad \xi_i \geq 0$$

$$\max \sum_{i=1}^m \alpha_i - \sum_{i=1}^m \sum_{j=1}^m y^{(i)} y^{(j)} \alpha_i \alpha_j \langle x^{(i)}, x^{(j)} \rangle \quad \text{s.t.} \quad \sum_{i=1}^m y^{(i)} \alpha_i = 0 \quad (0 \leq \alpha_i \leq C) \quad (i=1, \dots, m)$$

Protein sequence classifier 예시 문제 to show how to design kernel

A...Z $\phi(x) = ?$

AAAA
AAAB
:
zzzz

$$\begin{bmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 1 \end{bmatrix} = \phi(x)$$

$$20^4 = 160,000$$

$$\phi(x)^T \phi(z) = k(x, z)$$

KMP algorithm 2차 문제